Dmitrii **Altukhov**

Software Engineer · PhD (Computer Science)

Amsterdam, The Netherlands

dmalt | dmitrii-altukhov-258b05233 | altukhov.dm@gmail.com | +316 136 36 770

Profile

Software engineer with track record of building high-performance data processing algorithms and systems. PhD in Computer Science with strong quantitative foundation in mathematics and timeseries analysis. Combining mathematical modelling with practical implementation skills to deliver reliable, high-performance software solutions for complex quantitative problems.

Programming languages Python, Java, C++/C, Javascript (React, NodeJS)
Quantitative skills Algorithms and data structures, linear algebra, statistics, ML, digital signal processing
Tools & Technologies AWS, Kafka, Redis, SQL, PyTorch, Triton, Docker

Experience

Creative Fabrica — *Amsterdam, the Netherlands*

Python engineer (SupplyAI) — *Dec. 2023 – present*

- Redesigned the architecture for image generation backend, significantly improving development efficency and system performance.
- Developed advanced algorithm for color quantization that became foundational for industry-leading image vectorization pipeline.
- Designed and implemented automated code quality system for team's monorepo, improving reliability and development efficiency.
- Technologies stack: **Python**, **Triton**, **React**, **NodeJS**, **AWS**, **Kafka**, **Redis**.

SBDA Group — *Remote*

Data Scientist (contract) — *Sep. 2023 – Oct 2023*

- Designed and implemented an end-to-end algorithmic solution in Python for optimizing a distributed manufacturing process.

Artificial Intelligence Research Institute (AIRI) — *Moscow, Russia*

Research fellow — *Feb. 2022 – Apr. 2023*

- Built and trained deep learning model to predict speech from MEG data in **PyTorch**, meg_speech_decoding on GitHub.
- Developed real-time **Python** pipeline for EEG timeseries prediction using Kalman filtering.

Higher School of Economics, Centre for Cognition and Decision Making — *Moscow, Russia*

Junior research fellow — *Feb. 2015 – Dec. 2021*

- Led 4-people group on a project for real-time feature extraction and GPU-accelerated 3D visualization of brain activity in **Python**.
- Sped up by a factor of 10 the beamformer inverse solver in MNE-python by modifying the algorithm for vectorized computations.

University of Montreal, CERNEC lab. — *Montreal, Canada*

Visiting researcher, temporary position — *Oct. – Dec. 2015, May 2016 – Dec. 2016*

- Co-developed an open-source **Python** package for GPU-accelerated brain data visualization Visbrain, Combrisson et al. [2019].

Scientific Research Institute of System Analysis — *Moscow, Russia*

Junior research fellow — *Jun. 2011 – Jan. 2015*

- Developed a droplet evaporation model in **C++**; tested CUDA-accelerated cluster software for simulations of flow in jet engines.

Education

Ph.D. in Computer Science — *Moscow, Russia*

Higher School of Economics, Faculty of Computer Science — *Jan. 2016 – Nov. 2021*

GPA: 4.0; proposed an algorithmic framework for brain connectivity estimation; research published in the leading domain journals.

Specialist degree in Mechanics *(Masters equivalent)* — *Moscow, Russia*

Lomonosov Moscow State University, Department of Mechanics and Mathematics — *Sep. 2008 – Jun. 2013*

Thesis: "Enhancement and validation of LOGOS software for simulations of the reactive fluid flows."

Algorithms, Part 1 and Part 2 — *Online*

Coursera, Princeton University — *2020*

Comprehensive Algorithms and Data Structures course in **Java**.